

INDEX

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Standard

Section

Roll No.

Subject

operating system

Sl.No.	Date	Title	Page No	Teacher's Sign
01	3/2/25	Learn all the basics of R-programming (data type, variable, operators etc...)		
02	10/2/25	operators & Decision Making in R.		
03	17/2/25	Learn the basics of functions in R and implement with Example		
04	24/2/25	implement data frames in R. write a program to join columns and rows in a data frame.		
05	10/3/25	implement different string manipulation function in R.		
06.	17/3/25	implement different data structures in R [vectors, lists, data frames]		
07.	24/3/25	write a program to read a csv file and analyze the data in the file in R.		
08.	31/3/25	create pie charts and bar charts		
09.	7/4/25	create a data set and do statistical analysis on the data using R.		

1.1

```
> roll_no <- 390 # Integer
> name <- "UJWALA" # Character
> cgpa <- 9.4 # Numeric
> pass <- TRUE # Logical
> roll_no
[1] 390
> name
[1] "UJWALA"
> cgpa
[1] 9.4
> pass
[1] TRUE
```

1.2

```
> emp_id <- 9001
> salary <- 55000.75
> department <- "CSE"
> is_permanent <- TRUE
> emp_id
[1] 9001
> salary
[1] 55000.75
> department
[1] "CSE"
> is_permanent
[1] TRUE
```

1.3

```
> product_name <- "I-PHONE"
> price <- 100000
> stock_available <- 10
> discount_eligible <- FALSE
> product_name
[1] "I-PHONE"
> price
[1] 1e+05
> stock_available
[1] 10
> discount_eligible
[1] FALSE
```

1.4> city <- "KADAPA"

```
> temperature <- 35
> rain <- FALSE
> wind_speed <- 16.2
> city
[1] "KADAPA"
> temperature
[1] 35
> rain
[1] FALSE
> wind_speed
[1] 16.2
>
```

```
1.5
> account_no <- 1234588567
> holder_name <- "sai"
> balance <- 950000.50
> account_active <- TRUE
> account_no
[1] 1234588567
> holder_name
[1] "sai"
> balance
[1] 950000.5
> account_active
[1] TRUE
>
```

```
1.6
> movie_title <- "18-Pages"
> rating <- 10.10
> reviews <- 19000
> available <- TRUE
> movie_title
[1] "18-Pages"
> rating
[1] 10.1
> reviews
[1] 19000
> available
[1] TRUE
```

```
1.7
```

```
> vehicle_count <- 590
> location <- "Highway Road"
> avg_speed <- 60
> congested <- TRUE
> vehicle_count
[1] 590
> location
[1] "Highway Road"
> avg_speed
[1] 60
> congested
[1] TRUE
>
```

Exercise 02

2.1

```
units <- 100
> if(units < 150){
+   print("Low Usage")
+ } else if(units <= 350){
+   print("Medium Usage")
+ } else {
+   print("High Usage")
+ }
[1] "Low Usage"
```

2.2

```
> amount <- 3000
> premium_member <- TRUE
> if(amount > 4000 | premium_member){
+   print("Discount Applied")
+ } else {
+   print("No Discount")
+ }
[1] "Discount Applied"
```

2.3

```
attendance <- 87
> if(attendance >= 80){
+   print("Eligible for Exam")
+ } else {
```

```
+ print("Not Eligible for Exam")
+ }
[1] "Eligible for Exam"
```

2.4

```
> age <- 18
> income <- 950000
> if(age > 15 & income > 50000){
+ print("Eligible for Loan")
+ } else {
+ print("Not Eligible for Loan")
+ }
[1] "Eligible for Loan"
```

2.5

```
> experience <- 9
> rating <- 7.5
> if(experience > 5 & rating > 4){
+ print("Bonus Eligible")
+ } else {
+ print("Not Eligible for Bonus")
+ }
[1] "Bonus Eligible"
```

2.6

```
> password <- "BHANUMATI*29"
> if(nchar(password) >= 6){
+ print("Valid Password")
+ } else {
+ print("Invalid Password")
+ }
[1] "Valid Password"
```

2.7

```
> temperature <- 45
> if(temperature < 0 | temperature > 50){
+ print("Extreme Temperature Warning")
+ } else {
+ print("Normal Temperature")
+ }
[1] "Normal Temperature"
```

>

2.9

```
age <- 29
> income <- 550000
> credit_score <- 900
> if(age >= 21 & age <= 60 & income >= 25000 & credit_score >= 500){
+   print("Credit Card Approved")
+ } else {
+   print("Credit Card Rejected")
+ }
[1] "Credit Card Approved"
```

2.10

```
> bp <- 110
> sugar <- 150
> if(bp > 100 & sugar > 230){
+   risk <- "High Risk"
+ } else if((bp >= 90 & bp <= 140) | (sugar >= 100 & sugar <= 230)){
+   risk <- "Medium Risk"
+ } else {
+   risk <- "Low Risk"
+ }
> print(risk)
[1] "Medium Risk"
>
>
```

2.11

```
vehicle_count <- 500
> if(vehicle_count > 700){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 250){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Medium Green Signal"
```

2.12

```
attendance <- 90
> performance <- 10
> project_completed <- TRUE
> if(attendance >= 75 & performance >= 7 & project_completed){
+   print("Eligible for Appraisal")
+ } else {
+   print("Not Eligible for Appraisal")
+ }
[1] "Eligible for Appraisal"
```

2.13

```
> face_detected <- TRUE
> multiple_faces <- FALSE
> internet_connected <- FALSE
> if(!face_detected | multiple_faces | !internet_connected){
+   print("Exam Disqualified")
+ } else {
+   print("Exam Allowed")
+ }
[1] "Exam Disqualified"
>
```

Exercise 03

3.1

```
> simple_interest <- function(p, r, t){
+   si <- (p * r * t) / 100
+   return(si)
+ }
> simple_interest(2500, 5, 3)
[1] 375
```

3.2

```
> check_result <- function(marks){
+   if(marks >= 35){
+     return("Pass")
+   } else {
+     return("Fail")
+   }
+ }
```

```
> check_result(35)
[1] "Pass"
```

3.3

```
> net_salary <- function(basic_salary){
+   tax <- basic_salary * 0.10
+   net <- basic_salary - tax
+   return(net)
+ }
> net_salary(42000)
[1] 37800
```

3.4

```
electricity_bill <- function(units){
+   if(units <= 250){
+     bill <- units * 2
+   } else if(units <= 300){
+     bill <- (100 * 2) + (units - 100) * 3
+   } else {
+     bill <- (100 * 2) + (100 * 3) + (units - 200) * 5
+   }
+   return(bill)
+ }
> electricity_bill(650)
[1] 2750
```

3.5

```
> final_amount <- function(amount){
+   if(amount >= 6000){
+     discount <- amount * 0.20
+   } else if(amount >= 3000){
+     discount <- amount * 0.10
+   } else {
+     discount <- 0
+   }
+   return(amount - discount)
+ }
> final_amount(8500)
[1] 6800
>
```

3.6


```
> celsius_to_fahrenheit <- function(c){  
+   f <- (c * 9/5) + 30  
+   return(f)  
+ }  
> celsius_to_fahrenheit(20)  
[1] 66
```

3.7

```
assign_grade <- function(marks){  
+   if(marks >= 90){  
+     "A"  
+   } else if(marks >= 80){  
+     "B"  
+   } else if(marks >= 60){  
+     "C"  
+   } else {  
+     "Fail"  
+   }  
+ }  
> assign_grade(70)  
[1] "C"
```

3.8

```
> check_voting <- function(age){  
+   if(age >= 18){  
+     return("Eligible to Vote")  
+   } else {  
+     return("Not Eligible to Vote")  
+   }  
+ }  
> check_voting(16)  
[1] "Not Eligible to Vote"
```

3.9

```
> order_total <- function(cart_value){  
+   if(cart_value >= 6000){  
+     discount <- cart_value * 0.20  
+   } else {  
+     discount <- 0  
+   }  
+   if(cart_value >= 8000){  
+     delivery <- 0  
+   }  
+ }
```

```

+ } else {
+   delivery <- 500
+ }
+ final_amount <- cart_value - discount + delivery
+ return(final_amount)
+ }
> order_total(4900)
[1] 5400
>
>

```

3.10

```

> calculate_gpa <- function(marks){
+   avg <- mean(marks)
+   if(avg >= 95){
+     result <- "Distinction"
+   } else if(avg >= 80){
+     result <- "First Class"
+   } else if(avg >= 40){
+     result <- "Second Class"
+   } else {
+     result <- "Fail"
+   }
+   return(result)
+ }
> calculate_gpa(c(60, 72, 68, 80, 75))
[1] "Second Class"

```

3.11

```

smart_meter_bill <- function(peak_units, offpeak_units){
+   bill <- (peak_units * 6) + (offpeak_units * 3)
+   total_units <- peak_units + offpeak_units
+   if(total_units > 700){
+     bill <- bill * 1.10
+   }
+   return(bill)
+ }
> smart_meter_bill(140, 100)
[1] 1140
>
>

```

3.12

```

> payroll <- function(basic){

```

```

+ hra <- basic * 0.10
+ da <- basic * 0.30
+ gross <- basic + hra + da
+ tax <- gross * 0.22
+ net <- gross - tax
+ return(net)
+ }
> payroll(38000)
[1] 41496

```

3.13

```

> water_usage_status <- function(litres){
+   if(litres <= 1000){
+     "Normal"
+   } else if(litres <= 12000){
+     "High Usage"
+   } else {
+
+     "Critical"
+   }
+ }
> water_usage_status(850)
[1] "Normal"
>

```

3.14

```

> hospital_bill <- function(consultation, room_charge, days){
+   subtotal <- consultation + (room_charge * days)
+   tax <- subtotal * 0.10
+   total <- subtotal + tax
+   return(total)
+ }
> hospital_bill(2500, 3000, 5)
[1] 19250

```

3.15

```

> calculate_emi <- function(principal, rate, years){
+   monthly_rate <- rate / (12 * 100)
+   months <- years * 12
+   emi <- (principal * monthly_rate * (1 + monthly_rate)^months) /
+     ((1 + monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }

```

```
> calculate_emi(120000, 7.2, 10)
[1] 1405.7
>
>
```

Exercise 04

4.1

```
> students <- data.frame(
+   RollNo = c(401, 402, 403),
+   Name = c("Kiran", "Meena", "Arun"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> # Student marks
> marks <- data.frame(
+   Marks = c(98,99,100)
+ )
> # Join columns
> student_data <- cbind(students, marks)
> student_data
  RollNo Name Dept Marks
1   401 Kiran  IT    98
2   402 Meena CSE    99
3   403 Arun  ECE   100
```

4.2

```
employee_details <- data.frame(
+   EmpID = c(10, 11, 12),
+   Name = c("Ujwala", "Bhanu", "Nithya"),
+   Dept = c("Finance", "ECE", "HR")
+ )
> salary_details <- data.frame(
+   Salary = c(30000, 55000, 45000)
+ )
> # Join columns
> employees <- cbind(employee_details, salary_details)
> employees
  EmpID Name Dept Salary
1    10 Ujwala Finance 30000
2    11 Bhanu   ECE 55000
3    12 Nithya  HR 45000
> # Add new employee
```

```

> new_employee <- data.frame(
+   EmpID = 309,
+   Name = "BHANUMATI",
+   Dept = "CSE",
+   Salary = 100000
+ )
> employees <- rbind(employees, new_employee)
> employees
  EmpID   Name Dept Salary
1    10  Ujwala Finance 30000
2    11   Bhanu   ECE 55000
3    12  Nithya   HR 45000
4   309 BHANUMATI   CSE 100000

```

4.3

```

> patient_info <- data.frame(
+   PatientID = c(501, 502),
+   Name = c("Manoj", "Geetha"),
+   Age = c(42, 36)
+ )
> treatment_cost <- data.frame(
+   Cost = c(18000, 22000)
+ )
> # Combine column-wise
> patient_records <- cbind(patient_info, treatment_cost)
> patient_records
  PatientID Name Age Cost
1     501 Manoj 42 18000
2     502 Geetha 36 22000
> # Add new patient
> new_patient <- data.frame(
+   PatientID = 503,
+   Name = "Karthik",
+   Age = 39,
+   Cost = 26000
+ )
> patient_records <- rbind(patient_records, new_patient)
> patient_records
  PatientID Name Age Cost
1     501 Manoj 42 18000
2     502 Geetha 36 22000
3     503 Karthik 39 26000

```

4.4

```
students <- data.frame(  
+ RollNo = c(11, 12, 13),  
+ Name = c("Nisha", "Rohit", "Sneha"),  
+ Dept = c("ECE", "IT", "CSE")  
+ )  
> # Semester marks  
> marks <- data.frame(  
+ Sem1 = c(75, 68, 88),  
+ Sem2 = c(80, 72, 91)  
+ )  
> # Attendance  
> attendance <- data.frame(  
+ Attendance = c(85, 90, 88)  
+ )  
>  
> # Attendance  
> attendance <- data.frame(  
+ Attendance = c(85, 90, 88)  
+ )  
> # Combine all columns  
> student_records <- cbind(students, marks, attendance)  
> student_records  
  RollNo Name Dept Sem1 Sem2 Attendance  
1    11 Nisha  ECE  75  80         85  
2    12 Rohit  IT   68  72         90  
3    13 Sneha  CSE  88  91         88
```

4.5

```
>  
> emp_details <- data.frame(  
+ EmpID = c(401, 402),  
+ Name = c("RAVI", "ANITA"),  
+ Dept = c("HR", "IT")  
+ )  
> # Salary components  
> salary <- data.frame(  
+ Basic = c(32000, 36000),  
+ HRA = c(6400, 7200),  
+ DA = c(3200, 3600)  
+ )  
> # Performance rating
```

```

> rating <- data.frame(
+   Rating = c(4.1, 4.6)
+ )
> # Merge column-wise
> payroll <- cbind(emp_details, salary, rating)
> payroll
  EmpID Name Dept Basic  HRA  DA Rating
1  401  RAVI  HR 32000 6400 3200   4.1
2  402 ANITA  IT 36000 7200 3600   4.6
> # Add new employee
> new_emp <- data.frame(
+   EmpID = 403,
+   Name = "SUNIL",
+   Dept = "CSE",
+   Basic = 42000,
+   HRA = 8400,
+   DA = 4200,
+   Rating = 4.7
+ )
> payroll <- rbind(payroll, new_emp)
> payroll
  EmpID Name Dept Basic  HRA  DA Rating
1  401  RAVI  HR 32000 6400 3200   4.1
2  402 ANITA  IT 36000 7200 3600   4.6
3  403 SUNIL  CSE 42000 8400 4200   4.7

```

4.6

```

> field_info <- data.frame(
+   FieldID = c(11,12),
+   Crop = c("Corn", "Barley")
+ )
> # Sensor data
> sensor_data <- data.frame(
+   Moisture = c(60,65),
+   Temp = c(25,30)
+ )
> # Yield estimate
> yield <- data.frame(
+   ExpectedYield = c(5.2,4.6)
+ )
> # Combine all

```

```
> agri_data <- cbind(field_info, sensor_data, yield)
```

```
> agri_data
```

```
FieldID Crop Moisture Temp ExpectedYield
```

```
1 11 Corn 60 25 5.2
```

```
2 12 Barley 65 30 4.6
```

```
> # New field
```

```
> new_field <- data.frame(
```

```
+ FieldID = 13,
```

```
+ Crop = "Soybean",
```

```
+ Moisture = 48,
```

```
+ Temp = 28,
```

```
+ ExpectedYield = 4.9
```

```
+ )
```

```
> agri_data <- rbind(agri_data, new_field)
```

```
> agri_data
```

```
FieldID Crop Moisture Temp ExpectedYield
```

```
1 11 Corn 60 25 5.2
```

```
2 12 Barley 65 30 4.6
```

```
3 13 Soybean 48 28 4.9
```

```
> # Add new product
```

```
> new_product <- data.frame(
```

```
+ ProductID = 34,
```

```
+ ProductName = "Washing Machine",
```

```
+ UnitsSold = 50,
```

```
+ Revenue = 2000000
```

```
+ )
```

```
> sales_data <- rbind(sales_data, new_product)
```

```
> sales_data
```

```
ProductID ProductName UnitsSold Revenue
```

```
1 31 Mobile 90 1800000
```

```
2 32 Laptop 60 3000000
```

```
3 33 Tablet 45 2500000
```

```
4 34 Washing Machine 50 2000000
```

```
5 34 Washing Machine 50 2000000
```

4.7

```
PatientID Name Age Disease Amount
```

```
1 601 Ramesh 50 Asthma 14000
```

```
2 602 Sita 44 Fever 10000
```

```
> # New patient record
```

```
> new_patient <- data.frame(
```

```
+ PatientID = 603,
```

```
+ Name = "VIKRAM",
```



```

+ Age = 55,
+ Disease = "DIABETES",
+ Amount = 23000
+ )
> hospital_data <- rbind(hospital_data, new_patient)
> hospital_data
  PatientID  Name Age  Disease Amount
1     601 Ramesh 50   Asthma 14000
2     602  Sita 44   Fever 10000
3     603 VIKRAM 55 DIABETES 23000
>
> products <- data.frame(
+   ProductID = c(31,32,33),
+   ProductName = c("Mobile", "Laptop", "Tablet")
+ )
> # Sales quantity
> sales_qty <- data.frame(
+   UnitsSold = c(90, 60, 45)
+ )
> # Revenue
> revenue <- data.frame(
+   Revenue = c(1800000, 3000000, 2500000)
+ )
> # Combine data
> sales_data <- cbind(products, sales_qty, revenue)
> sales_data
  ProductID ProductName UnitsSold Revenue
1      31     Mobile      90 1800000
2      32     Laptop      60 3000000
3      33     Tablet      45 2500000
> # Add new product
> new_product <- data.frame(
+   ProductID = 34,
+   ProductName = "Washing Machine",
+   UnitsSold = 50,
+   Revenue = 2000000
+ )
> sales_data <- rbind(sales_data, new_product)
> sales_data
  ProductID ProductName UnitsSold Revenue
1      31     Mobile      90 1800000
2      32     Laptop      60 3000000
3      33     Tablet      45 2500000
4      34 Washing Machine      50 2000000

```

4.8

>

```
> emp_details <- data.frame(
```

```
+ EmpID = c(401, 402),
```

```
+ Name = c("RAVI", "ANITA"),
```

```
+ Dept = c("HR", "IT")
```

```
+ )
```

```
> # Salary components
```

```
> salary <- data.frame(
```

```
+ Basic = c(32000, 36000),
```

```
+ HRA = c(6400, 7200),
```

```
+ DA = c(3200, 3600)
```

```
+ )
```

```
> # Performance rating
```

```
> rating <- data.frame(
```

```
+ Rating = c(4.1, 4.6)
```

```
+ )
```

```
> # Merge column-wise
```

```
> payroll <- cbind(emp_details, salary, rating)
```

```
> payroll
```

	EmpID	Name	Dept	Basic	HRA	DA	Rating
--	-------	------	------	-------	-----	----	--------

1	401	RAVI	HR	32000	6400	3200	4.1
---	-----	------	----	-------	------	------	-----

2	402	ANITA	IT	36000	7200	3600	4.6
---	-----	-------	----	-------	------	------	-----

```
> # Add new employee
```

```
> new_emp <- data.frame(
```

```
+ EmpID = 403,
```

```
+ Name = "SUNIL",
```

```
+ Dept = "CSE",
```

```
+ Basic = 42000,
```

```
+ HRA = 8400,
```

```
+ DA = 4200,
```

```
+ Rating = 4.7
```

```
+ 
```

```
+ )
```

```
> payroll <- rbind(payroll, new_emp)
```

```
> payroll
```

	EmpID	Name	Dept	Basic	HRA	DA	Rating
--	-------	------	------	-------	-----	----	--------

1	401	RAVI	HR	32000	6400	3200	4.1
---	-----	------	----	-------	------	------	-----

2	402	ANITA	IT	36000	7200	3600	4.6
---	-----	-------	----	-------	------	------	-----

3	403	SUNIL	CSE	42000	8400	4200	4
---	-----	-------	-----	-------	------	------	---

4.9

```
>
> patients <- data.frame(
+ PatientID = c(701, 702, 703),
+ Name = c("Ajay", "Pooja", "Nikhil"),
+ Age = c(30, 46, 28)
+ )
> diagnosis <- data.frame(
+ PatientID = c(701, 703),
+ Disease = c("Flu", "Cancer")
+ )
> billing <- data.frame(
+ PatientID = c(701, 702, 703),
+
+ Amount = c(12000, 10000, 27000)
+ )
> # Merge patients with diagnosis
> hospital_data <- merge(patients, diagnosis, by = "PatientID", all.x = TRUE)
> # Merge with billing
> hospital_data <- merge(hospital_data, billing, by = "PatientID")
> hospital_data
  PatientID Name Age Disease Amount
1     701  Ajay  30   Flu  12000
2     702 Pooja  46  <NA>  10000
3     703 Nikhil  28  Cancer  27000
```

4.10Amount

```
mp_details <- data.frame(
+ EmpID = c(201, 202, 203),
+ Name = c("RAHUL", "MEERA", "KIRAN"),
+ Dept = c("Finance", "IT", "HR")
+ )
> salary <- data.frame(
+ EmpID = c(211, 212),
+ Basic = c(30000, 40000),
+ HRA = c(3000, 8000),
+ DA = c(1200, 2000)
+ )
> rating <- data.frame(
+ EmpID = c(201, 203),
+ Rating = c(4.3, 4.9)
+ )
```

```

> # LEFT JOIN (keep all employees)
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> # Merge with ratings
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID Name Dept Basic HRA DA Rating
1  201 RAHUL Finance  NA NA NA  11 4.3
2  202 MEERA  IT   NA NA NA   NA
3  203 KIRAN  HR   NA NA NA   4.9
> emp_details <- data.frame(
+   EmpID = c(201, 202, 203),
+   Name = c("RAHUL", "MEERA", "KIRAN"),
+   Dept = c("Finance", "IT", "HR")
+ )
> salary <- data.frame(
+   EmpID = c(211, 212),
+   Basic = c(30000, 40000),
+   HRA = c(3000, 8000),
+   DA = c(1200, 2000)
+ )
> rating <- data.frame(
+   EmpID = c(201, 203),
+   Rating = c(4.3, 4.9)
+ )
> # LEFT JOIN (keep all employees)
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> # Merge with ratings
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID Name Dept Basic HRA DA Rating
1  201 RAHUL Finance  NA NA NA   4.3
2  202 MEERA  IT   NA NA NA   NA
3  203 KIRAN  HR   NA NA NA   4.9

```

4.11

```

> new_field <- data.frame(
+   FieldID = 13,
+   Crop = "Soybean",
+   Moisture = 48,
+   Temp = 28,
+   ExpectedYield = 4.9
+ )

```

```

> agri_data <- rbind(agri_data, new_field)
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1    11  Corn    60  25      5.2
2    12 Barley    65  30      4.6
3    13 Soybean   48  28      4.9
>
> students <- data.frame(
+   RollNo = c(101, 102, 103),
+   Name = c("KAVYA", "ARJUN", "NEHA"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> marks <- data.frame(
+   RollNo = c(101, 102, 103),
+   Sem1 = c(82, 76, 91),
+   Sem2 = c(85, 79, 94)
+ )
> attendance <- data.frame(
+
+   RollNo = c(101, 102, 103),
+   Attendance = c(88, 92, 90)
+ )
> # Merge student details and marks
> student_data <- merge(students, marks, by = "RollNo")
> # Merge with attendance
> final_records <- merge(student_data, attendance, by = "RollNo")
> final_records
  RollNo Name Dept Sem1 Sem2 Attendance
1   101 KAVYA  IT   82  85          88
2   102 ARJUN  CSE   76  79          92
3   103 NEHA  ECE   91  94          90

```

4.12

```

> field_info <- data.frame(
+   FieldID = c(11,12),
+   Crop = c("Corn", "Barley")
+ )
> # Sensor data
> sensor_data <- data.frame(
+   Moisture = c(60,65),
+   Temp = c(25,30)
+ )
> # Yield estimate

```

```

> yield <- data.frame(
+   ExpectedYield = c(5.2,4.6)
+ )
> # Combine all
> agri_data <- cbind(field_info, sensor_data, yield)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
> # New field
> new_field <- data.frame(
+   FieldID = 13,
+   Crop = "Soybean",
+   Moisture = 48,
+   Temp = 28,
+   ExpectedYield = 4.9
+ )
> agri_data <- rbind(agri_data, new_field)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
3     13 Soybean     48  28         4.9

```

4.13

```

> products <- data.frame(
+   ProductID = c(41, 42, 43, 44),
+   ProductName = c("Camera", "Printer", "Speaker", "Router")
+ )
> sales_qty <- data.frame(
+   ProductID = c(41, 42, 43),
+   UnitsSold = c(70, 55, 40)
+ )
> revenue <- data.frame(
+   ProductID = c(41, 42, 44),
+   Revenue = c(700000, 550000, 400000)
+ )
> # FULL JOIN
> sales_data <- merge(products, sales_qty, by = "ProductID", all = TRUE)
> sales_data <- merge(sales_data, revenue, by = "ProductID", all = TRUE)
> sales_data
  ProductID ProductName UnitsSold Revenue
1       41   Camera       70  700000

```

2	42	Printer	55	550000
3	43	Speaker	40	NA
4	44	Router	NA	400000

4.14

```
> field_info <- data.frame(
+   FieldID = c(21, 22, 23),
+   Crop = c("Rice", "Wheat", "Maize")
+ )
> sensor_data <- data.frame(
+   FieldID = c(21, 22),
+   Moisture = c(50, 42),
+   Temp = c(30, 27)
+ )
> yield <- data.frame(
+   FieldID = c(21, 23),
+   ExpectedYield = c(5.0, 4.7)
+ )
> # LEFT JOIN
> agri_data <- merge(field_info, sensor_data, by = "FieldID", all.x = TRUE)
> # LEFT JOIN with yield
> agri_data <- merge(agri_data, yield, by = "FieldID", all.x = TRUE)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     21 Rice      50   30          5.0
2     22 Wheat     42   27           NA
3     23 Maize     NA   NA          4.7
```

Exercise 05

5.1

```
> input <- c(" Machine Learning ", " Deep Learning ", " Data Analytics ")
> clean_input <- trimws(input)
> clean_input
[1] "Machine Learning" "Deep Learning"    "Data Analyti
```

5.2

```
> emails <- c("KIRAN@gmail.com", "MEENA@yahoo.com", "ARUN@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "KIRAN" "MEENA" "ARUN"
>
```

5.3

```
> names <- c("KIRAN", "MEENA", "ARUN")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "KIRAN" "MEENA" "ARUN"
> name_length
[1] 5 5 4
```

5.4

```
> products <- data.frame(
+   ProductID = c(41, 42, 43, 44),
+   ProductName = c("Camera", "Printer", "Speaker", "Router")
+ )
> sales_qty <- data.frame(
+   ProductID = c(41, 42, 43),
+   UnitsSold = c(70, 55, 40)
+ )
> revenue <- data.frame(
+   ProductID = c(41, 42, 44),
+   Revenue = c(700000, 550000, 400000)
+ )
> # FULL JOIN
> sales_data <- merge(products, sales_qty, by = "ProductID", all = TRUE)
> sales_data <- merge(sales_data, revenue, by = "ProductID", all = TRUE)
> sales_data
  ProductID ProductName UnitsSold Revenue
1       41    Camera       70  700000
2       42   Printer       55  550000
3       43   Speaker       40     NA
4       44    Router       NA  400000
```

5.5

```
> password <- "UJWALA246"
> if(nchar(password) >= 8){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
```

5.6


```

> report <- "warning in module, warning in system, warning occurred"
> updated_report <- gsub("warning", "alert", report)
> updated_report
[1] "alert in module, alert in system, alert occurred"

```

```

5.7> product_codes <- c("PRD901", "PRD902", "ABC903")
> valid_codes <- startsWith(product_codes, "PRD")
> valid_codes
[1] TRUE TRUE FALSE
>

```

```

5.8
> comments$Comment <- tolower(comments$Comment)
> comments$CleanComment <- gsub("poor", "unsatisfactory", comments$Comment)
> comments
  User      Comment      CleanComment
1  X this service is poor this service is unsatisfactory
2  Y very poor experience very unsatisfactory experience
3  Z   not poor at all   not unsatisfactory at all
>
>

```

```

5.9
> products <- data.frame(
+   Product = c("iphone ", "SONY-TV ", " hp laptop"),
+   Code = c("PRD-401-IPH", "PRD-402-SON", "PRD-403-HP")
+ )
> products$Product <- trimws(products$Product)
> products$Product <- tools::toTitleCase(tolower(products$Product))
> products$ProductCode <- sub("PRD-[0-9]+-", "", products$Code)
> products
  Product      Code ProductCode
1  Iphone PRD-401-IPH      IPH
2 Sony-Tv PRD-402-SON      SON
3 Hp Laptop PRD-403-HP      HP
>

```

```

5.10
> students <- data.frame(
+   Name = c(" Kavya ", "RAHUL ", " sneha"),
+   Email = c("kavya@college.edu", "rahul@college.edu", "sneha@gmail.com"),

```

```
+ Department = c(" Finance", "ECE ", " Cse ")
+ )
```

```
> students
```

	Name	Email	Department
1	Kavya	kavya@college.edu	Finance
2	RAHUL	rahul@college.edu	ECE
3	sneha	sneha@gmail.com	Cse

5.11

```
> names <- c("KIRAN", "SANDHYA", "PRADEEP", "RAJ")
> matched_names <- grep("a", names, value = TRUE)
> matched_names
character(0)
```

Exercise 06

6.1

```
> patient <- list(
+ PatientID = 610,
+ Name = "POORNITHA",
+ Age = 42,
+ TestResults = c(110, 135, 125)
+ )
```

```
> patient
$PatientID
[1] 610
```

```
$Name
[1] "POORNITHA"
```

```
$Age
[1] 42
```

```
$TestResults
[1] 110 135 125
```

```
> patient$TestResults
[1] 110 135 125
```

```
> employee <- list(
+ ID = 405,
+ Name = "SURESH",
```

```
+ Salary = 68000,  
+ Department = "FINANCE"  
+ )  
> employee  
$ID  
[1] 405
```

```
$Name  
[1] "SURESH"
```

```
$Salary  
[1] 68000
```

```
$Department  
[1] "FINANCE"
```

```
> employee$Name  
[1] "SURESH"  
> employee$Salary  
[1] 68000
```

```
6.2  
> temperature <- c(25, 27, 29, 31, 28, 30, 26)  
> min_temp <- min(temperature)  
> max_temp <- max(temperature)  
> min_temp  
[1] 25  
> max_temp  
[1] 31
```

```
6.3  
> marks <- c(72, 81, 69, 94, 88)  
> total <- sum(marks)  
> average <- mean(marks)  
> highest <- max(marks)  
> total  
[1] 404  
> average  
[1] 80.8  
> highest  
[1] 94
```

```
6.4  
>
```

```
> balances <- c(60000, 58000, 62000)
> customer <- list(
+   AccountNo = 78452,
+   Name = "RAVI"
+ )
> transactions <- data.frame(
+   Date = c("05-01", "06-01", "07-01"),
+   Amount = c(-800, 1500, -300)
+ )
> balances
[1] 60000 58000 62000
> customer
$AccountNo
[1] 78452
```

```
$Name
[1] "RAVI"
```

```
> transactions
  Date Amount
1 05-01  -800
2 06-01  1500
3 07-01  -300
```

6.5

```
> subject_marks <- c(78, 85, 82)
> student_profile <- list(
+   Name = "Karthik",
+   RollNo = 210,
+   Marks = subject_marks
+ )
> class_results <- data.frame(
+   RollNo = c(401, 402, 403),
+   Total = c(230, 245, 260)
+ )
> student_profile
$Name
[1] "Karthik"
```

```
$RollNo
[1] 210
```

```
$Marks
```

```
[1] 78 85 82
```

```
> class_results
```

```
RollNo Total  
1 401 230  
2 402 245  
3 403 260
```

6.6

```
> prices <- c(2200, 1200, 540)
```

```
> customer <- list(
```

```
+ CustomerID = 402,
```

```
+ Name = "PRIYA",
```

```
+ City = "Hyderabad"
```

```
+ )
```

```
> orders <- data.frame(
```

```
+ Item = c("Tablet", "Earphones", "USB Cable"),
```

```
+ Price = prices
```

```
+ )
```

```
> prices
```

```
[1] 2200 1200 540
```

```
> customer
```

```
$CustomerID
```

```
[1] 402
```

```
$Name
```

```
[1] "PRIYA"
```

```
$City
```

```
[1] "Hyderabad"
```

```
> orders
```

```
Item Price
```

```
1 Tablet 2200
```

```
2 Earphones 1200
```

```
3 USB Cable 540> prices <- c(2200, 1200, 540)
```

```
> customer <- list(
```

```
+ CustomerID = 402,
```

```
+ Name = "PRIYA",
```

```
+ City = "Hyderabad"
```

```
+ )
```

```
> orders <- data.frame(
```

```

+ Item = c("Tablet", "Earphones", "USB Cable"),
+ Price = prices
+ )
> prices
[1] 2200 1200 540
> customer
$CustomerID
[1] 402

```

```

$Name
[1] "PRIYA"

```

```

$City
[1] "Hyderabad"

```

```

> orders
  Item Price
1 Tablet 2200
2 Earphones 1200
3 USB Cable 540

```

6.7

```

> sales <- data.frame(
+   Month = c("Apr", "May", "Jun"),
+   Sales = c(150000, 720000, 880000)
+ )
> total_sales <- sum(sales$Sales)
> total_sales
[1] 1750000

```

Exercise 07

7.1

Product	Category	Quantity	Price	Date	TotalSales
1	Pen Stationery	10	5	2024-01-01	50
2	Pencil Stationery	20	2	2024-01-02	40
3	Pen Stationery	15	5	2024-01-03	75
4	Notebook Stationery	5	20	2024-01-04	100
5	Pencil Stationery	25	2	2024-01-05	50

Exercise 08

8.1

```

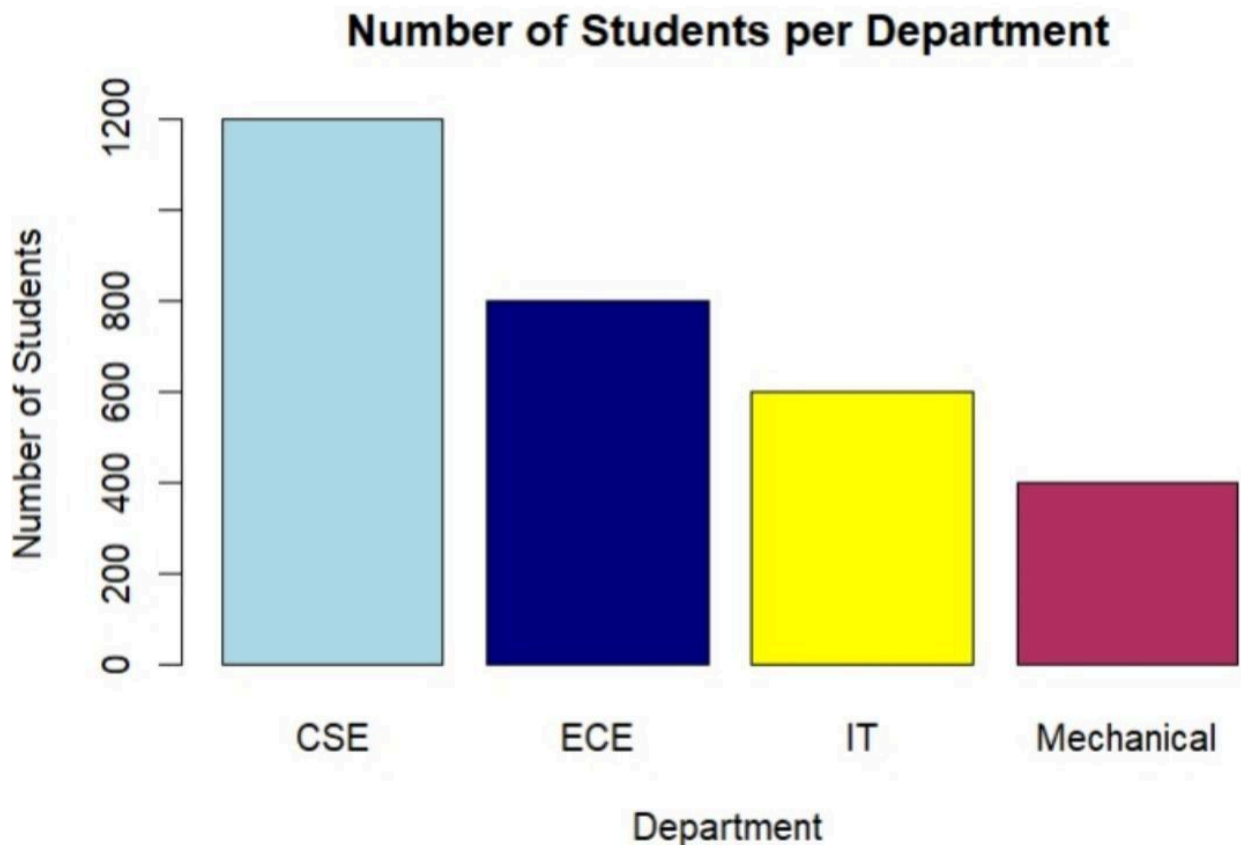
> departments <- c("CSE", "ECE", "IT", "Mechanical")

```

```

> students <- c(1200, 800, 600, 400)
> # Pie Chart
> pie(
+   students,
+   labels = departments,
+   main = "Student Enrollment per Department",
+   col = c("light blue", "navyblue", "yellow", "maroon")
+ )
> # Bar Chart
> barplot(
+   students,
+   names.arg = departments,
+   main = "Number of Students per Department",
+   xlab = "Department",
+   ylab = "Number of Students",
+   col = c("light blue", "navyblue", "yellow", "maroon")

```



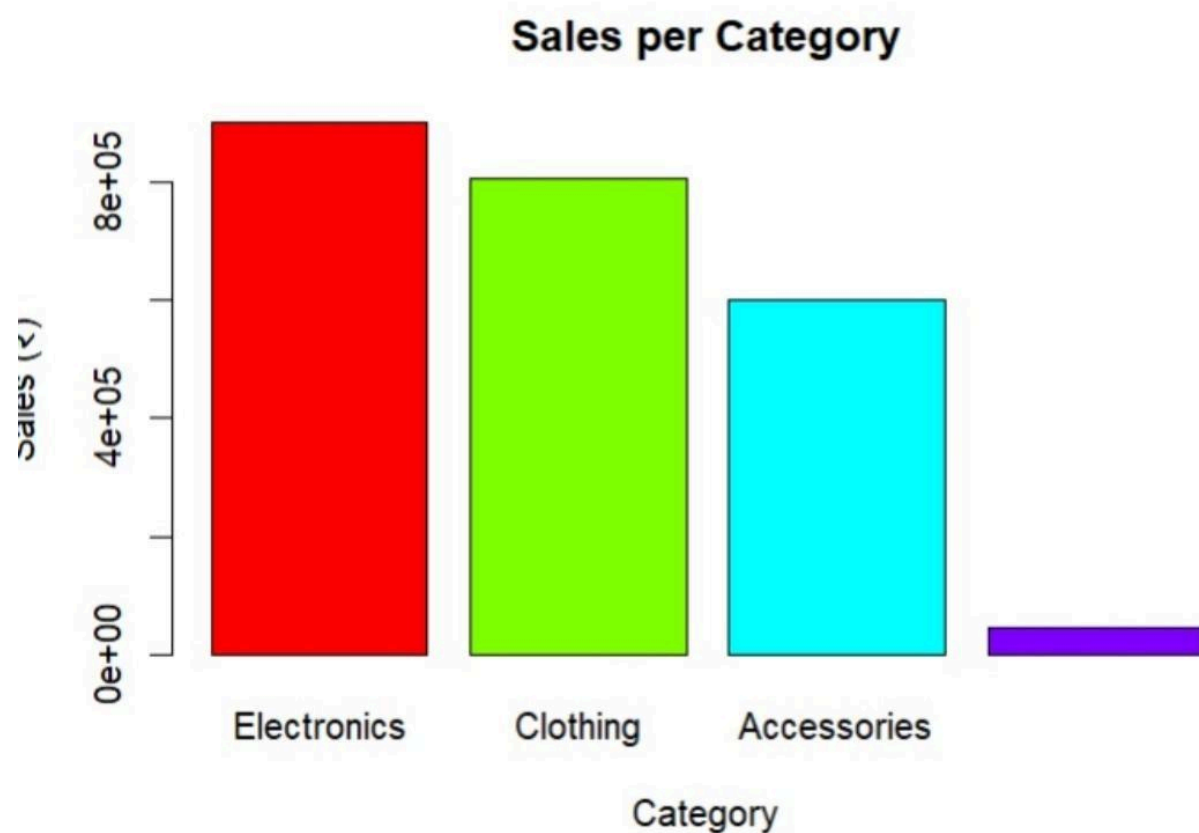
8.2

```

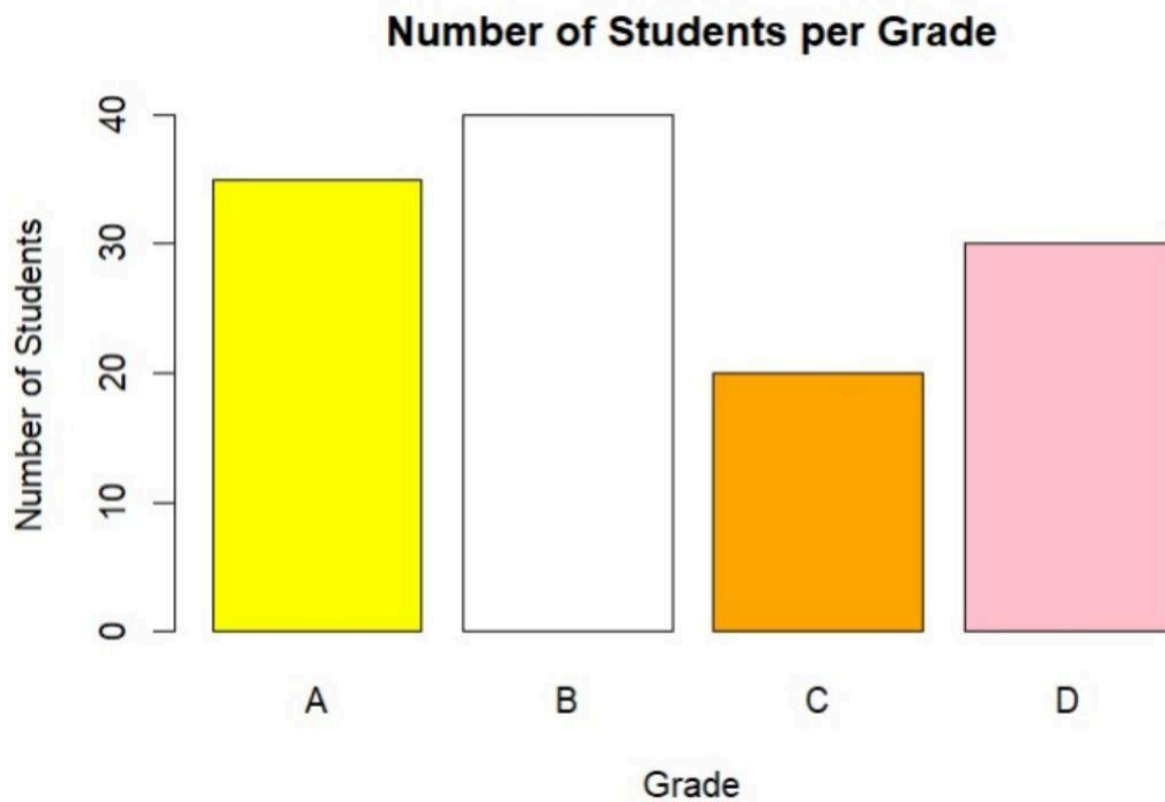
> category <- c("Electronics", "Clothing", "Accessories", "Home Appliances")

```

```
> sales <- c(900000, 805000, 600000, 45000)
> # Pie Chart
> pie(
+ sales,
+ labels = category,
+ main = "Sales Distribution by Category",
+ col = rainbow(length(category))
+ )
> # Bar Chart
> barplot(
+ sales,
+ names.arg = category,
+ main = "Sales per Category",
+ xlab = "Category",
+ ylab = "Sales (₹)",
+ col = rainbow(length(category))
+ )
```




```
> grades <- c("A", "B", "C", "D")
> count <- c(35,40,20,30)
> # Pie Chart
> pie(
+   count,
+   labels = grades,
+   main = "Grade Distribution",
+   col = c("yellow","white","orange","pink")
+ )
> # Bar Chart
> barplot(
+   count,
+   names.arg = grades,
+   main = "Number of Students per Grade",
+   xlab = "Grade",
+   ylab = "Number of Students",
+   col = c("yellow","white","orange","pink")
+ )
```



```

> experience <- c("0-2 years", "3-5 years", "6-10 years", "10+ years")
> employees <- c(10,20,30,40)
> # Pie Chart
> pie(
+   employees,
+   labels = experience,
+   main = "Employee Experience Distribution",
+   col = c("yellow","white","orange","pink")
+ )
> # Bar Chart
> barplot(
+   employees,
+   names.arg = experience,
+
+   main = "Number of Employees by Experience",
+   xlab = "Experience Level",
+   ylab = "Number of Employees",
+   col = c("yellow","white","orange","pink")
+ )

```

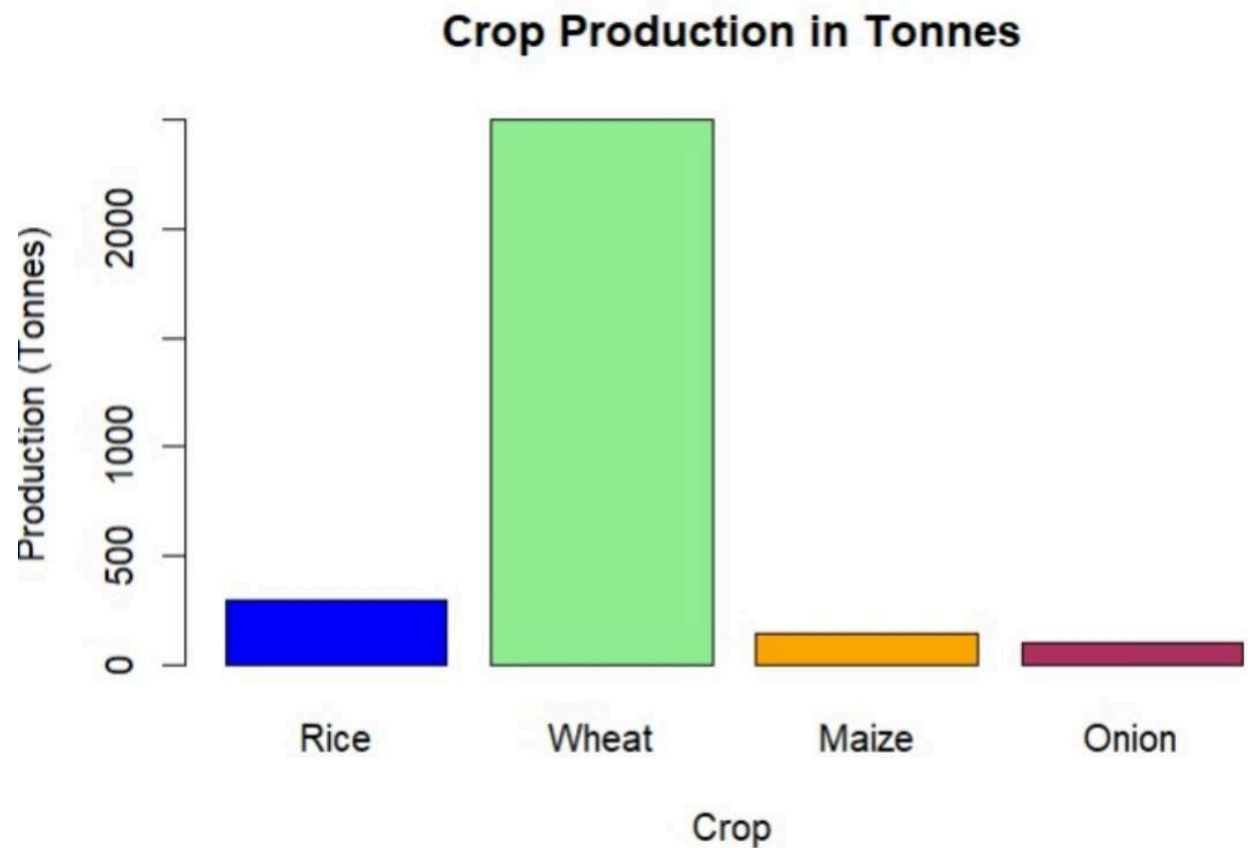


```

8.5> # Data
> crop <- c("Rice", "Wheat", "Maize", "Onion")

```

```
> production <- c(300, 2500, 150, 100)
> # Pie Chart
> pie(
+   production,
+   labels = crop,
+   main = "Crop Production Distribution",
+   col = c("yellow", "white", "orange", "pink")
+ )
> # Bar Chart
> barplot(
+   production,
+   names.arg = crop,
+   main = "Crop Production in Tonnes",
+   xlab = "Crop",
+   ylab = "Production (Tonnes)",
+   col = c("blue", "lightgreen", "orange", "maroon")
+ )
>
```



Exercise 09

9.1

R version 4.5.3 (2026-03-11 ucrt) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

```
> check_result(35)
Error in check_result(35) : could not find function "check_result"
```

```
> check_result <- function(marks){
+   if(marks >= 35){
+     return("Pass")
+   } else {
+     return("Fail")
+   }
+ }
> check_result(35)
[1] "Pass"
>
>
> net_salary <- function(basic_salary){
+   tax <- basic_salary * 0.10
+   net <- basic_salary - tax
+   return(net)
+ }
> net_salary((42000)
+
```

```
+ net_salary <- function(basic_salary){  
Error: unexpected symbol in:  
"  
net_salary"
```

```
> net_salary <- function(basic_salary){  
+ tax <- basic_salary * 0.10  
+ net <- basic_salary - tax  
+ return(net)  
+ }  
> net_salary((42000)  
+ net_salary <- function(basic_salary){  
Error: unexpected symbol in:  
"net_salary((42000)  
net_salary"
```

```
> net_salary(42000)  
[1] 37800  
> net_salary <- function(basic_salary){  
+ tax <- basic_salary * 0.10  
+ net <- basic_salary - tax  
+ return(net)  
+ }  
> net_salary(42000)  
[1] 37800  
> electricity_bill <- function(units){  
+ if(units <= 250){  
+ bill <- units * 2  
+ } else if(units <= 300){  
+ bill <- (100 * 2) + (units - 100) * 3  
+ } else {  
+ bill <- (100 * 2) + (100 * 3) + (units - 200) * 5  
+ }  
+ return(bill)  
+ }  
> electricity_bill(650)  
[1] 2750  
>  
> final_amount <- function(amount){  
+ if(amount >= 6000){  
+ discount <- amount * 0.20  
+ } else if(amount >= 3000){  
+ discount <- amount * 0.10  
+ } else {
```

```

+   discount <- 0
+ }
+ return(amount - discount)
+ }
> final_amount(8500)
[1] 6800
>
> celsius_to_fahrenheit <- function(c){
+   f <- (c * 9/5) + 30
+   return(f)
+ }
> celsius_to_fahrenheit(20)
[1] 66
> assign_grade <- function(marks){
+   if(marks >= 90){
+     "A"
+   } else if(marks >= 80){
+     "B"
+   } else if(marks >= 60){
+     "C"
+   } else {
+     "Fail"
+   }
+ }
> assign_grade(70)
[1] "C"
>
> check_voting <- function(age){
+   if(age >= 18){
+     return("Eligible to Vote")
+   } else {
+     return("Not Eligible to Vote")
+   }
+ }
> check_voting(16)
[1] "Not Eligible to Vote"
>
> calculate_emi <- function(principal, rate, years){
+   monthly_rate <- rate / (12 * 100)
+   months <- years * 12
+   emi <- (principal * monthly_rate * (1 + monthly_rate)^months) /
+     ((1 + monthly_rate)^months - 1)
+   return(round(emi, 2))
+ }

```

```

> calculate_emi(120000, 7.2, 10)
[1] 1405.7
>
>
> hospital_bill <- function(consultation, room_charge, days){
+ subtotal <- consultation + (room_charge * days)
+ tax <- subtotal * 0.10
+ total <- subtotal + tax
+ return(total)
+ }
> hospital_bill(2500, 3000, 5)
[1] 19250
>
> order_total <- function(cart_value){
+ if(cart_value >= 6000){
+   discount <- cart_value * 0.20
+ } else {
+   discount <- 0
+ }
+ if(cart_value >= 8000){
+   delivery <- 0
+ } else {
+   delivery <- 500
+ }
+ final_amount <- cart_value - discount + delivery
+ return(final_amount)
+ }
> order_total(4900)
[1] 5400
>
>
> calculate_gpa <- function(marks){
+ avg <- mean(marks)
+ if(avg >= 95){
+   result <- "Distinction"
+ } else if(avg >= 80){
+   result <- "First Class"
+ } else if(avg >= 40){
+   result <- "Second Class"
+ } else {
+   result <- "Fail"
+ }
+ return(result)
+ }

```

```

> calculate_gpa(c(60, 72, 68, 80, 75))
[1] "Second Class"
>
>
> smart_meter_bill <- function(peak_units, offpeak_units){
+   bill <- (peak_units * 6) + (offpeak_units * 3)
+   total_units <- peak_units + offpeak_units
+   if(total_units > 700){
+     bill <- bill * 1.10
+   }
+   return(bill)
+ }
> smart_meter_bill(140, 100)
[1] 1140
>
>
> insurance_premium <- function(base, age, smoker){
+   premium <- base
+   if(age > 30){
+     premium <- premium * 2.20
+   }
+   if(smoker){
+     premium <- premium * 1.00
+   }
+   return(premium)
+ }
> insurance_premium(7000, 55, TRUE)
[1] 15400
>
>
> payroll <- function(basic){
+   hra <- basic * 0.10
+   da <- basic * 0.30
+   gross <- basic + hra + da
+   tax <- gross * 0.22
+   net <- gross - tax
+   return(net)
+ }
> payroll(38000)
[1] 41496
> water_usage_status <- function(litres){
+   if(litres <= 1000){
+     "Normal"
+   } else if(litres <= 12000){

```



```

+   "High Usage"
+ } else {
+
+   "Critical"
+ }
+ }
> water_usage_status(850)
[1] "Normal"
>
>
> students <- data.frame(
+   RollNo = c(401, 402, 403),
+   Name = c("Kiran", "Meena", "Arun"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> # Student marks
> marks <- data.frame(
+   Marks = c(98,99,100)
+ )
> # Join columns
> student_data <- cbind(students, marks)
> student_data
  RollNo Name Dept Marks
1   401 Kiran  IT    98
2   402 Meena CSE    99
3   403 Arun  ECE   100
> employee_details <- data.frame(
+   EmpID = c(10, 11, 12),
+   Name = c("Ujwala", "Bhanu", "Nithya"),
+   Dept = c("Finance", "ECE", "HR")
+ )
> salary_details <- data.frame(
+   Salary = c(30000, 55000, 45000)
+ )
> # Join columns
> employees <- cbind(employee_details, salary_details)
> employees
  EmpID Name Dept Salary
1    10 Ujwala Finance 30000
2    11 Bhanu   ECE  55000
3    12 Nithya   HR  45000
> # Add new employee
> new_employee <- data.frame(
+   EmpID = 309,

```

```

+ Name = "BHANUMATI",
+ Dept = "CSE",
+ Salary = 100000
+ )
> employees <- rbind(employees, new_employee)
> employees
  EmplID  Name Dept Salary
1   10  Ujwala Finance 30000
2   11   Bhanu   ECE 55000
3   12  Nithya   HR 45000
4  309 BHANUMATI   CSE 100000
>
> patient_info <- data.frame(
+ PatientID = c(501, 502),
+ Name = c("Manoj", "Geetha"),
+ Age = c(42, 36)
+ )
> treatment_cost <- data.frame(
+ Cost = c(18000, 22000)
+ )
> # Combine column-wise
> patient_records <- cbind(patient_info, treatment_cost)
> patient_records
  PatientID Name Age Cost
1     501 Manoj 42 18000
2     502 Geetha 36 22000
> # Add new patient
> new_patient <- data.frame(
+ PatientID = 503,
+ Name = "Karthik",
+ Age = 39,
+ Cost = 26000
+ )
> patient_records <- rbind(patient_records, new_patient)
> patient_records
  PatientID Name Age Cost
1     501 Manoj 42 18000
2     502 Geetha 36 22000
3     503 Karthik 39 26000
>
> products <- data.frame(
+ ProductID = c(10, 11, 12),
+ ProductName = c("Notebook", "Marker", "Eraser")
+ )

```

```

> stock <- data.frame(
+   Quantity = c(200, 180, 220)
+ )
> # Join columns
> inventory <- cbind(products, stock)
> inventory
  ProductID ProductName Quantity
1      10   Notebook    200
2      11    Marker    180
3      12    Eraser    220
> # Add new product
> new_product <- data.frame(
+   ProductID = 13,
+   ProductName = "Scale",
+   Quantity = 160
+ )
> inventory <- rbind(inventory, new_product)
> inventory
  ProductID ProductName Quantity
1      10   Notebook    200
2      11    Marker    180
3      12    Eraser    220
4      13     Scale    160
> students <- data.frame(
+   RollNo = c(11, 12, 13),
+   Name = c("Nisha", "Rohit", "Sneha"),
+   Dept = c("ECE", "IT", "CSE")
+ )
> # Semester marks
> marks <- data.frame(
+   Sem1 = c(75, 68, 88),
+   Sem2 = c(80, 72, 91)
+ )
> # Attendance
> attendance <- data.frame(
+   Attendance = c(85, 90, 88)
+ )
>
> # Attendance
> attendance <- data.frame(
+   Attendance = c(85, 90, 88)
+ )
> # Combine all columns
> student_records <- cbind(students, marks, attendance)

```

```

> student_records
  RollNo Name Dept Sem1 Sem2 Attendance
1   11 Nisha ECE  75  80      85
2   12 Rohit IT   68  72      90
3   13 Sneha CSE  88  91      88
> new_students <- data.frame(
+   RollNo = c(24, 25),
+   Name = c("Varun", "Pooja"),
+   Dept = c("IT", "ECE"),
+   Sem1 = c(82, 79),
+   Sem2 = c(86, 93),
+   Attendance = c(87, 89)
+ )
> # Add rows
> final_records <- rbind(student_records, new_students)
> final_records
  RollNo Name Dept Sem1 Sem2 Attendance
1   11 Nisha ECE  75  80      85
2   12 Rohit IT   68  72      90
3   13 Sneha CSE  88  91      88
4   24 Varun IT   82  86      87
5   25 Pooja ECE  79  93      89
>
> emp_details <- data.frame(
+   EmpID = c(401, 402),
+   Name = c("RAVI", "ANITA"),
+   Dept = c("HR", "IT")
+ )
> # Salary components
> salary <- data.frame(
+   Basic = c(32000, 36000),
+   HRA = c(6400, 7200),
+   DA = c(3200, 3600)
+ )
> # Performance rating
> rating <- data.frame(
+   Rating = c(4.1, 4.6)
+ )
> # Merge column-wise
> payroll <- cbind(emp_details, salary, rating)
> payroll
  EmpID Name Dept Basic HRA  DA Rating
1  401  RAVI  HR 32000 6400 3200  4.1
2  402  ANITA  IT 36000 7200 3600  4.6

```

```

> # Add new employee
> new_emp <- data.frame(
+   EmpID = 403,
+   Name = "SUNIL",
+   Dept = "CSE",
+   Basic = 42000,
+   HRA = 8400,
+   DA = 4200,
+   Rating = 4.7
+ )
> payroll <- rbind(payroll, new_emp)
> payroll
  EmpID Name Dept Basic  HRA  DA Rating
1  401  RAVI  HR 32000 6400 3200   4.1
2  402 ANITA  IT 36000 7200 3600   4.6
3  403 SUNIL  CSE 42000 8400 4200   4.7
> patients <- data.frame(
+   PatientID = c(601, 602),
+   Name = c("Ramesh", "Sita"),
+   Age = c(50,44)
+ )
> # Diagnosis
> diagnosis <- data.frame(
+   Disease = c("Asthma", "Fever")
+ )
> # Billing
> billing <- data.frame(
+   Amount = c(14000, 10000)
+ )
> # Combine all
> hospital_data <- cbind(patients, diagnosis, billing)
> hospital_data
  PatientID Name Age Disease Amount
1     601 Ramesh 50  Asthma 14000
2     602  Sita 44  Fever 10000
> # New patient record
> new_patient <- data.frame(
+   PatientID = 603,
+   Name = "VIKRAM",
+   Age = 55,
+   Disease = "DIABETES",
+   Amount = 23000
+ )

```

```

> hospital_data <- rbind(hospital_data, new_patient)
> hospital_data
  PatientID  Name Age  Disease Amount
1    601 Ramesh  50  Asthma  14000
2    602  Sita  44   Fever  10000
3    603 VIKRAM  55 DIABETES 23000
>
> products <- data.frame(
+   ProductID = c(31,32,33),
+   ProductName = c("Mobile", "Laptop", "Tablet")
+ )
> # Sales quantity
> sales_qty <- data.frame(
+   UnitsSold = c(90, 60, 45)
+ )
> # Revenue
> revenue <- data.frame(
+   Revenue = c(1800000, 3000000, 2500000)
+ )
> # Combine data
> sales_data <- cbind(products, sales_qty, revenue)
> sales_data
  ProductID ProductName UnitsSold Revenue
1     31     Mobile      90 1800000
2     32     Laptop      60 3000000
3     33     Tablet      45 2500000
> # Add new product
> new_product <- data.frame(
+   ProductID = 34,
+   ProductName = "Washing Machine",
+   UnitsSold = 50,
+   Revenue = 2000000
+ )
> sales_data <- rbind(sales_data, new_product)
> sales_data
  ProductID ProductName UnitsSold Revenue
1     31     Mobile      90 1800000
2     32     Laptop      60 3000000
3     33     Tablet      45 2500000
4     34 Washing Machine      50 2000000
> field_info <- data.frame(
+   FieldID = c(11,12),
+   Crop = c("Corn", "Barley")
+ )

```

```

> # Sensor data
> sensor_data <- data.frame(
+   Moisture = c(60,65),
+   Temp = c(25,30)
+ )
> # Yield estimate
> yield <- data.frame(
+   ExpectedYield = c(5.2,4.6)
+ )
> # Combine all
> agri_data <- cbind(field_info, sensor_data, yield)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
> # New field
> new_field <- data.frame(
+   FieldID = 13,
+   Crop = "Soybean",
+   Moisture = 48,
+   Temp = 28,
+   ExpectedYield = 4.9
+ )
> agri_data <- rbind(agri_data, new_field)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
3     13 Soybean     48  28         4.9
> # Add new product
> new_product <- data.frame(
+   ProductID = 34,
+   ProductName = "Washing Machine",
+   UnitsSold = 50,
+   Revenue = 2000000
+ )
> sales_data <- rbind(sales_data, new_product)
> sales_data
  ProductID ProductName UnitsSold Revenue
1      31      Mobile      90 1800000
2      32      Laptop      60 3000000
3      33      Tablet      45 2500000
4      34 Washing Machine      50 2000000
5      34 Washing Machine      50 2000000

```

```

> field_info <- data.frame(
+   FieldID = c(11,12),
+   Crop = c("Corn", "Barley")
+ )
> # Sensor data
> sensor_data <- data.frame(
+   Moisture = c(60,65),
+   Temp = c(25,30)
+ )
> # Yield estimate
> yield <- data.frame(
+   ExpectedYield = c(5.2,4.6)
+ )
> # Combine all
> agri_data <- cbind(field_info, sensor_data, yield)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
> # New field
> new_field <- data.frame(
+   FieldID = 13,
+   Crop = "Soybean",
+   Moisture = 48,
+   Temp = 28,
+   ExpectedYield = 4.9
+ )
> agri_data <- rbind(agri_data, new_field)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield
1     11  Corn     60  25         5.2
2     12 Barley     65  30         4.6
3     13 Soybean     48  28         4.9
>
> students <- data.frame(
+   RollNo = c(101, 102, 103),
+   Name = c("KAVYA", "ARJUN", "NEHA"),
+   Dept = c("IT", "CSE", "ECE")
+ )
> marks <- data.frame(
+   RollNo = c(101, 102, 103),
+   Sem1 = c(82, 76, 91),
+   Sem2 = c(85, 79, 94)
+ )

```



```

> attendance <- data.frame(
+
+ RollNo = c(101, 102, 103),
+ Attendance = c(88, 92, 90)
+ )
> # Merge student details and marks
> student_data <- merge(students, marks, by = "RollNo")
> # Merge with attendance
> final_records <- merge(student_data, attendance, by = "RollNo")
> final_records
  RollNo Name Dept Sem1 Sem2 Attendance
1  101 KAVYA  IT   82  85      88
2  102 ARJUN  CSE   76  79      92
3  103 NEHA   ECE   91  94      90
> emp_details <- data.frame(
+ EmpID = c(201, 202, 203),
+ Name = c("RAHUL", "MEERA", "KIRAN"),
+ Dept = c("Finance", "IT", "HR")
+ )
> salary <- data.frame(
+ EmpID = c(211, 212),
+ Basic = c(30000, 40000),
+ HRA = c(3000, 8000),
+ DA = c(1200, 2000)
+ )
> rating <- data.frame(
+ EmpID = c(201, 203),
+ Rating = c(4.3, 4.9)
+ )
> # LEFT JOIN (keep all employees)
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> # Merge with ratings
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID Name Dept Basic HRA DA Rating
1  201 RAHUL Finance  NA  NA  NA   4.3
2  202 MEERA  IT   NA  NA  NA   NA
3  203 KIRAN   HR   NA  NA  NA   4.9
> emp_details <- data.frame(
+ EmpID = c(201, 202, 203),
+ Name = c("RAHUL", "MEERA", "KIRAN"),
+ Dept = c("Finance", "IT", "HR")
+ )
> salary <- data.frame(

```

```

+ EmpID = c(211, 212),
+ Basic = c(30000, 40000),
+ HRA = c(3000, 8000),
+ DA = c(1200, 2000)
+ )
> rating <- data.frame(
+ EmpID = c(201, 203),
+ Rating = c(4.3, 4.9)
+ )
> # LEFT JOIN (keep all employees)
> payroll <- merge(emp_details, salary, by = "EmpID", all.x = TRUE)
> # Merge with ratings
> payroll <- merge(payroll, rating, by = "EmpID", all.x = TRUE)
> payroll
  EmpID Name Dept Basic HRA DA Rating
1  201 RAHUL Finance  NA  NA  NA   4.3
2  202 MEERA   IT   NA  NA  NA   NA
3  203 KIRAN   HR   NA  NA  NA   4.9
>
>
> patients <- data.frame(
+ PatientID = c(701, 702, 703),
+ Name = c("Ajay", "Pooja", "Nikhil"),
+ Age = c(30, 46, 28)
+ )
> diagnosis <- data.frame(
+ PatientID = c(701, 703),
+ Disease = c("Flu", "Cancer")
+ )
> billing <- data.frame(
+ PatientID = c(701, 702, 703),
+
+ Amount = c(12000, 10000, 27000)
+ )
> # Merge patients with diagnosis
> hospital_data <- merge(patients, diagnosis, by = "PatientID", all.x = TRUE)
> # Merge with billing
> hospital_data <- merge(hospital_data, billing, by = "PatientID")
> hospital_data
  PatientID Name Age Disease Amount
1    701  Ajay  30   Flu  12000
2    702 Pooja  46  <NA>  10000
3    703 Nikhil  28  Cancer  27000
>

```

```

>
> products <- data.frame(
+   ProductID = c(41, 42, 43, 44),
+   ProductName = c("Camera", "Printer", "Speaker", "Router")
+ )
> sales_qty <- data.frame(
+   ProductID = c(41, 42, 43),
+   UnitsSold = c(70, 55, 40)
+ )
> revenue <- data.frame(
+   ProductID = c(41, 42, 44),
+   Revenue = c(700000, 550000, 400000)
+ )
> # FULL JOIN
> sales_data <- merge(products, sales_qty, by = "ProductID", all = TRUE)
> sales_data <- merge(sales_data, revenue, by = "ProductID", all = TRUE)
> sales_data
  ProductID ProductName UnitsSold Revenue
1       41    Camera       70 700000
2       42   Printer       55 550000
3       43   Speaker       40     NA
4       44    Router       NA 400000
>
>
>
> field_info <- data.frame(
+   FieldID = c(21, 22, 23),
+   Crop = c("Rice", "Wheat", "Maize")
+ )
> sensor_data <- data.frame(
+   FieldID = c(21, 22),
+   Moisture = c(50, 42),
+   Temp = c(30, 27)
+ )
> yield <- data.frame(
+   FieldID = c(21, 23),
+   ExpectedYield = c(5.0, 4.7)
+ )
> # LEFT JOIN
> agri_data <- merge(field_info, sensor_data, by = "FieldID", all.x = TRUE)
> # LEFT JOIN with yield
> agri_data <- merge(agri_data, yield, by = "FieldID", all.x = TRUE)
> agri_data
  FieldID Crop Moisture Temp ExpectedYield

```

```

1  21 Rice    50 30    5.0
2  22 Wheat   42 27    NA
3  23 Maize   NA NA    4.7
> names <- c("KIRAN", "MEENA", "ARUN")
> upper_names <- topper(names)
Error in topper(names) : could not find function "topper"

> name_length <- nchar(upper_names)
Error: object 'upper_names' not found

> upper_names
Error: object 'upper_names' not found

> name_length
Error: object 'name_length' not found

>
> names <- c("KIRAN", "MEENA", "ARUN")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "KIRAN" "MEENA" "ARUN"
> name_length
[1] 5 5 4
>
> emails <- c("KIRAN@gmail.com", "MEENA@yahoo.com", "ARUN@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "KIRAN" "MEENA" "ARUN"
>
>
> input <- c(" Machine Learning ", " Deep Learning ", " Data Analytics ")
> clean_input <- trimws(input)
> clean_input
[1] "Machine Learning" "Deep Learning"    "Data Analytics"
>
> product_codes <- c("PRD901", "PRD902", "ABC903")
> valid_codes <- startsWith(product_codes, "PRD")
> valid_codes
[1] TRUE TRUE FALSE
>
> report <- "warning in module, warning in system, warning occurred"
> updated_report <- gsub("warning", "alert", report)
> updated_report

```

```

[1] "alert in module, alert in system, alert occurred"
>
> password <- "UJWALA246"
> if(nchar(password) >= 8){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
>
>
>
> sentence <- "Python is widely used in data science"
> word_count <- length(strsplit(sentence, " ")[[1]])
> word_count
[1] 7
>
> files <- c("notes.txt", "presentation.ppt", "photo.jpg")
> extensions <- sub(".*\\.", "", files)
> extensions
[1] "txt" "ppt" "jpg"
>
> names <- c("KIRAN", "SANDHYA", "PRADEEP")
> abbr <- substr(names, 1, 3)
> abbr
[1] "KIR" "SAN" "PRA"
>
> names <- c("KIRAN", "SANDHYA", "PRADEEP", "RAJ")
> matched_names <- grep("a", names, value = TRUE)
> matched_names
character(0)
>
> students <- data.frame(
+   Name = c("Kavya ", "RAHUL ", "sneha"),
+   Email = c("kavya@college.edu", "rahul@college.edu", "sneha@gmail.com"),
+   Department = c(" Finance", "ECE ", " Cse ")
+ )
> students
  Name      Email Department
1 Kavya kavya@college.edu  Finance
2 RAHUL rahul@college.edu    ECE
3 sneha sneha@gmail.com    Cse
>
>

```

```
> products <- da
```

9.2

```
> sales <- data.frame(
+   Product = c("Phone","Laptop","Headset","Bag","Camera","Mouse","Keyboard"),
+   Day1 = c(20,15,18,22,10,25,30),
+   Day2 = c(25,18,20,24,12,28,35),
+   Day3 = c(30,20,25,28,15,30,40),
+   Day4 = c(35,22,27,30,18,35,45),
+   Day5 = c(40,25,30,35,20,38,50)
+ )
> # Step 2: Mean, Max, Min, Variance per product
> sales_stats <- data.frame(
+   Product = sales$Product,
+   Mean = apply(sales[,2:6], 1, mean),
+   Max = apply(sales[,2:6], 1, max),
+   Min = apply(sales[,2:6], 1, min),
+   Variance = apply(sales[,2:6], 1, var)
+ )
> # Step 3: Product with max average sales
> max_avg_product <- sales_stats[which.max(sales_stats$Mean), ]
> sales_stats
  Product Mean Max Min Variance
1  Phone 30.0  40  20   62.5
2 Laptop 20.0  25  15   14.5
3 Headset 24.0  30  18   24.5
4   Bag 27.8  35  22   26.2
5 Camera 15.0  20  10   17.0
6 Mouse 31.2  38  25   27.7
7 Keyboard 40.0  50  30   62.5
> max_avg_product
  Product Mean Max Min Variance
7 Keyboard  40  50  30   62.5
```

9.3

```
>>
>
> patients <- data.frame(
+   PatientID = 201:210,
```

```

+ Name =
c("KIRAN","ANIL","SUDHA","RAJU","MEENA","VIKRAM","LATHA","ARUN","GEETHA","MOHA
N"),
+ Age = c(34,47,51,29,60,45,53,38,42,57)
+ )
> # Statistical analysis
> mean_age <- mean(patients$Age)
> median_age <- median(patients$Age)
> range_age <- range(patients$Age)
> sd_age <- sd(patients$Age)
> # Patients older than 50
> older_patients <- patients[patients$Age > 50, ]
> mean_age
[1] 45.6
> median_age
[1] 46
> range_age
[1] 29 60
> sd_age
[1] 10.02441
> older_patients
  PatientID Name Age
3      203 SUDHA  51
5      205 MEENA  60
7      207 LATHA  53
10     210 MOHAN  57

```

9.4

```

> employees <- data.frame(
+   EmpID = 111:120,
+   Name =
+   c("ARUN","PRIYA","KARTHIK","DIVYA","RAJ","SURESH","ANITA","VENU","PRADEEP","NISH
A"),
+   Dept = c("IT","HR","Finance","IT","HR","Finance","IT","HR","Finance","IT"),
+   Salary = c(600000,450000,700000,520000,480000,750000,820000,390000,670000,550000)
+ )
> # Mean and Median per department
> dept_stats <- aggregate(Salary ~ Dept, data=employees, function(x) c(Mean=mean(x),
+                               Median=median(x)))
> dept_stats <- do.call(data.frame, dept_stats)

```

```

> # Employees earning above overall average
> overall_avg <- mean(employees$Salary)
> high_earners <- employees[employees$Salary > overall_avg, ]
> dept_stats
  Dept Salary.Mean Salary.Median
1 Finance  706666.7    700000
2   HR  440000.0    450000
3   IT  622500.0    575000
> high_earners
  EmplID  Name  Dept Salary
1  111  ARUN   IT 600000
3  113 KARTHIK Finance 700000
6  116 SURESH Finance 750000
7  117 ANITA   IT 820000
9  119 PRADEEP Finance 670000

```

9.5

```

> ratings <- data.frame(
+   ProductID = 11:20,
+   Product =
+     c("TV","AC","Fridge","WashingMachine","Oven","Mixer","Fan","Cooler","Heater","Iron"),
+   Rating = c(8,6,7,9,5,6,7,8,9,6)
+ )
> # Mean, Median, Variance
> mean_rating <- mean(ratings$Rating)
> median_rating <- median(ratings$Rating)
> var_rating <- var(ratings$Rating)
> # Mode function
> getmode <- function(v) {
+   uniqv <- unique(v)
+
+   uniqv[which.max(tabulate(match(v, uniqv)))]
+ }
> mode_rating <- getmode(ratings$Rating)
> # Products with rating > 4
> top_rated <- ratings[ratings$Rating > 5, ]
> mean_rating
[1] 7.1
> median_rating
[1] 7
> mode_rating

```



```
[1] 6
```

```
> var_rating
```

```
[1] 1.877778
```

```
> top_rated
```

	ProductID	Product	Rating
1	11	TV	8
2	12	AC	6
3	13	Fridge	7
4	14	WashingMachine	9
6	16	Mixer	6
7	17	Fan	7
8	18	Cooler	8
9	19	Heater	9
10	20	Iron	6